

Detailed versus global legalize

Global legalize lite maps to Tetris shelf pack, displacement six, HPWL thirty-two

The idea

- Pick global Tetris when you want a fast pass and can afford extra movement
- Pick detailed Abacus when displacement budget is tight
- Report both pipelines side by side in regressions, legal first, then disp and HPWL

Pseudocode

- Here the sketch is two named pipelines on the same seed
- Global legalize lite means Tetris
- Detailed legalize lite means Abacus
- Pseudocode names both so regressions do not mix the labels
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Both pipelines must end legal
- Then compare displacement six versus four and HPWL thirty-two versus thirty-eight
- Pick detailed when movement budget is tight; pick global when speed matters more

Algorithm sketch — try these

```
INPUT: illegal / global positions
global:   TetrisLite → disp=6 HPWL=32
detailed: AbacusLite → disp=4 HPWL=38
both must report legal=true
CHOOSE detailed when disp budget is tight
CHOOSE global when a fast shelf pack is enough
```

Global = Tetris, disp 6

DETAILED VS GLOBAL LEGALIZE

Global = Tetris, disp 6

Step 1 / 5 · global-tetris · module04-01-detailed-vs-global

chip 12×6 · 3 rows · site 1



Explanation

Global legalize lite maps to Tetris-style nearest-row shelf pack. On the overlap seed it legalizes with displacement six and HPWL thirty-two.

- globalLegalize → tetris
- single-row shelf for A,B,C
- fast, simple pipeline stage

```
globalDisp: 6  
HPWL: 32  
legal: true
```

Global = Tetris, disp 6

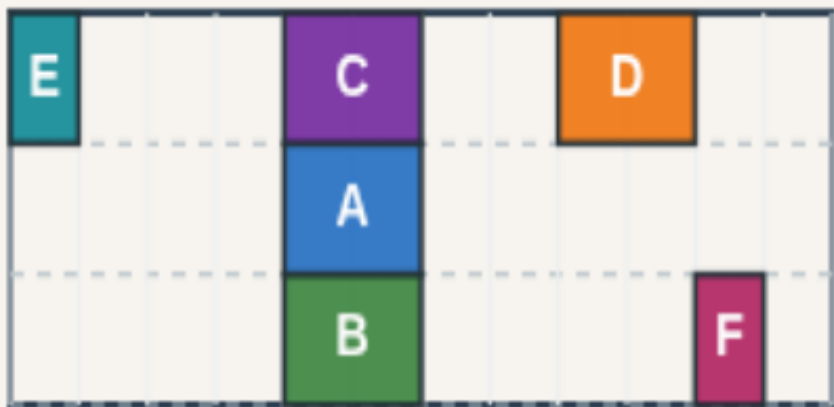
Detailed = Abacus, disp 4

DETAILED VS GLOBAL LEGALIZE

Detailed = Abacus, disp 4

Step 2 / 5 · detailed-abacus · module04-01-detailed-vs-global

chip 12×6 · 3 rows · site 1



Explanation

Detailed legalize lite maps to Abacus row trial. Same seed, both legal—but displacement drops to four by spreading A, B, C across rows.

- detailedLegalize → abacus
- cross-row assignment
- lower L1 movement

```
detailedDisp: 4  
HPWL: 38  
legal: true
```

Detailed = Abacus, disp 4

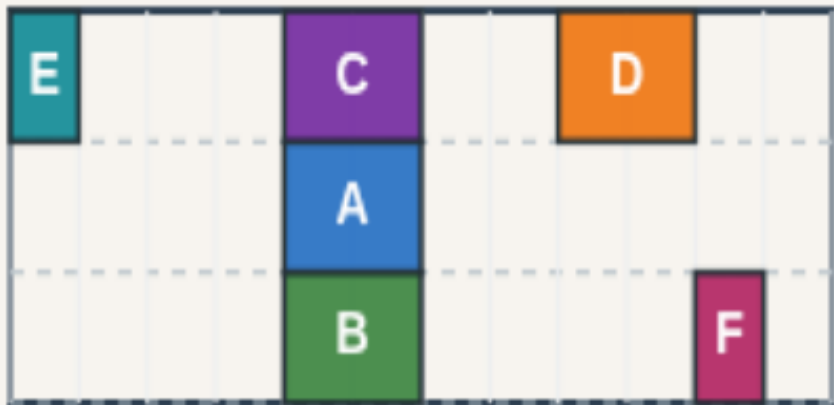
Both pipelines legal

DETAILED VS GLOBAL LEGALIZE

Both pipelines legal

Step 3 / 5 · both-legal · module04-01-detailed-vs-global

chip 12x6 · 3 rows · site 1



Explanation

Global and detailed both pass legality on the overlap seed. The difference is how far cells move and how HPWL shifts—thirty-two versus thirty-eight here.

- Same starter coordinates
- Both isLegal true
- Metrics tell the story

```
global legal: true
detailed legal: true
HPWL delta: 6
```

Both pipelines legal

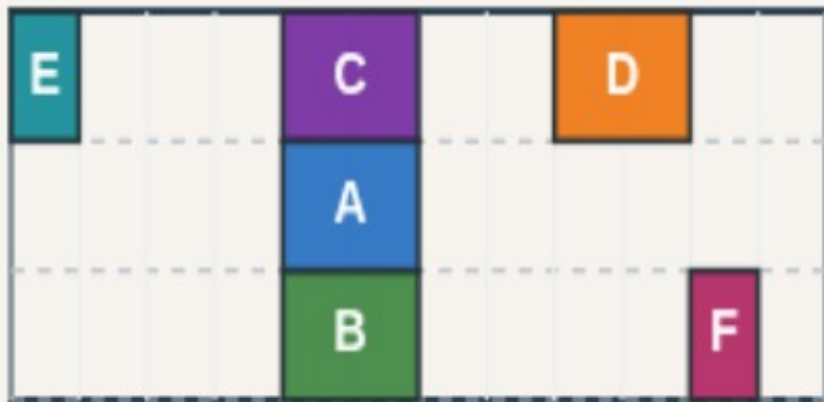
Side-by-side metrics

DETAILED VS GLOBAL LEGALIZE

Side-by-side metrics

Step 4 / 5 · side-by-side · module04-01-detailed-vs-global

chip 12×6 · 3 rows · site 1



Explanation

Global Tetris: disp six, HPWL thirty-two. Detailed Abacus: disp four, HPWL thirty-eight. Neither dominates on both axes—pick by your displacement budget.

- global: disp 6 HPWL 32
- detailed: disp 4 HPWL 38
- Report both in regressions

```
globalDisp: 6
detailedDisp: 4
tetrisHpwl: 32
abacusHpwl: 38
```

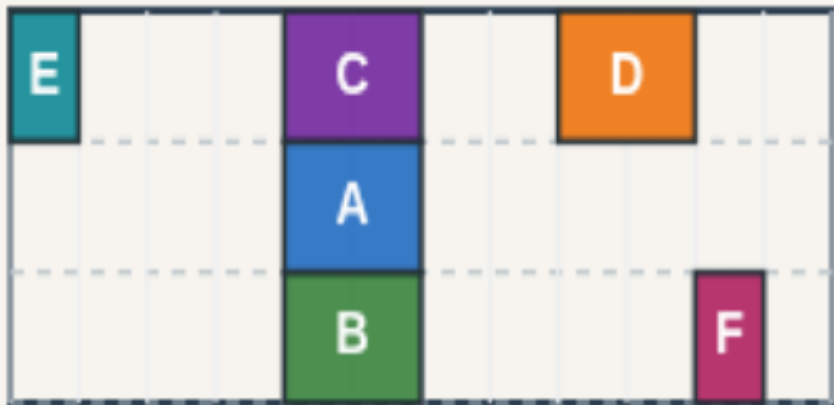
Pick detailed when displacement is tight

DETAILED VS GLOBAL LEGALIZE

Pick detailed when displacement is tight

Step 5 / 5 · pick-detailed · module04-01-detailed-vs-global

chip 12×6 · 3 rows · site 1



Explanation

Production flows often run a fast global legalize, then a detailed pass when timing or continuity needs cells near global targets. On this toy, that is Abacus over Tetris.

- Tight disp budget → detailed
- HPWL-first → global Tetris
- Course wrap: offline compare next

```
detailedDisp: 4  
globalDisp: 6  
both legal on overlap seed
```

Pick detailed when displacement is tight

Browser lab track

- In the browser lab track, open the **detailed-vs-global** lab from the tools shelf
- Open the interactive lab
- Reveal golden is study-only
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny\_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module